

Assignment 4

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G :

$$S \rightarrow aB,$$

$$S' \rightarrow S + B\varepsilon,$$

$$B \rightarrow \varepsilon.$$

G' :

$$S' \rightarrow S,$$

$$S \rightarrow aB,$$

$$B \rightarrow S + B|\varepsilon.$$

1 Bottom-Up Parsing Exercises

Exercise 1

1、Calculate the *FIRST* and *FOLLOW* sets for G .

$$FIRST(S) = \{a\},$$

$$FIRST(B) = \{a, \varepsilon\},$$

$$FOLLOW(S) = \{+, \$\},$$

$$FOLLOW(B) = \{+, \$\}.$$

Construct the collection of sets of *LR(0)* items for G' .

$I_0:$ $S' \rightarrow \cdot S$ $S \rightarrow \cdot aB$	$I_2:$ $S \rightarrow a \cdot B$ $B \rightarrow \cdot S + B$ $B \rightarrow \cdot$ $S \rightarrow \cdot aB$ $I_3:$ $S \rightarrow aB \cdot$	$I_5:$ $B \rightarrow S + \cdot B$ $B \rightarrow \cdot S + B$ $B \rightarrow \cdot$ $S \rightarrow \cdot aB$
$I_1:$ $S' \rightarrow S \cdot$	$I_4:$ $B \rightarrow S \cdot + B$	$I_6:$ $B \rightarrow S + B \cdot$

Construct the *SLR(1)* parsing table for G' .

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s2	r3	r3	4	3

3		r1	r1		
4		s5			
5	s2	r3	r3	4	6
6		r2	r2		

2、Yes, it is.

3、Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	shift
(2)	0 2	a	aaa + + + \$	shift
(3)	0 2 2	aa	aa + + + \$	shift
(4)	0 2 2 2	aaa	a + + + \$	shift
(5)	0 2 2 2 2	aaaa	+ + + \$	reduce by $B \rightarrow \varepsilon$
(6)	0 2 2 2 2 3	aaaaB	+ + + \$	reduce by $S \rightarrow aB$
(7)	0 2 2 2 4	aaaS	+ + + \$	shift
(8)	0 2 2 2 4 5	aaaS +	+ + \$	reduce by $B \rightarrow \varepsilon$
(9)	0 2 2 2 4 5 6	aaaS + B	+ + \$	reduce by $B \rightarrow S + B$
(10)	0 2 2 2 3	aaaB	+ + \$	reduce by $S \rightarrow aB$
(11)	0 2 2 4	aaS	+ + \$	shift
(12)	0 2 2 4 5	aaS +	+ \$	reduce by $B \rightarrow \varepsilon$
(13)	0 2 2 4 5 6	aaS + B	+ \$	reduce by $B \rightarrow S + B$
(14)	0 2 2 3	aaB	+ \$	reduce by $S \rightarrow aB$
(15)	0 2 4	aS	+ \$	shift
(16)	0 2 4 5	aS +	\$	reduce by $B \rightarrow \varepsilon$
(17)	0 2 4 5 6	aS + B	\$	reduce by $B \rightarrow S + B$
(18)	0 2 3	aB	\$	reduce by $S \rightarrow aB$
(19)	0 1	S	\$	shift

Exercise 2

1、Construct the collection of sets of $LR(0)$ items for G' .

$I_0:$ $S' \rightarrow \cdot S, \$$ $S \rightarrow \cdot aB, \$$	$I_2:$ $S \rightarrow a \cdot B, \$$ $B \rightarrow \cdot S + B, \$$ $B \rightarrow \cdot, \$$	$I_3:$ $S \rightarrow aB \cdot, \$$	$I_5:$ $B \rightarrow S + \cdot B, \$$ $B \rightarrow \cdot S + B, \$$ $B \rightarrow \cdot, \$$
$I_1:$ $S' \rightarrow S \cdot, \$$	$I_4:$ $S \rightarrow \cdot aB, +$	$I_4:$ $B \rightarrow S \cdot + B, \$$	$I_5:$ $S \rightarrow \cdot aB, +$

$I_6:$ $B \rightarrow S + B \cdot, \$$	$I_8:$ $S \rightarrow aB \cdot, +$	$I_{10}:$ $B \rightarrow S + \cdot B, +$ $B \rightarrow \cdot S + B, +$ $B \rightarrow \cdot, +$ $S \rightarrow \cdot aB, +$
$I_7:$ $S \rightarrow a \cdot B, +$ $B \rightarrow \cdot S + B, +$ $B \rightarrow \cdot, +$ $S \rightarrow \cdot aB, +$	$I_9:$ $B \rightarrow S \cdot + B, +$	$I_{11}:$ $B \rightarrow S + B \cdot, +$

Construct the canonical LR(1) parsing table for G' .

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s7		r3	4	3
3			r1		
4		s5			
5	s7		r3	4	6
6			r2		
7	s7	r3		9	8
8		r1			
9		s10			
10	s7	r3		9	11
11		r2			

2、Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	shift
(2)	0 2	a	aaa + + + \$	shift
(3)	0 2 7	aa	aa + + + \$	shift
(4)	0 2 7 7	aaa	a + + + \$	shift
(5)	0 2 7 7 7	aaaa	+ + + \$	reduce by $B \rightarrow \epsilon$
(6)	0 2 7 7 7 8	aaaaB	+ + + \$	reduce by $S \rightarrow aB$
(7)	0 2 7 7 9	aaaS	+ + + \$	shift
(8)	0 2 7 7 9 10	aaaS +	+ + \$	reduce by $B \rightarrow \epsilon$
(9)	0 2 7 7 9 10 11	aaaS + B	+ + \$	reduce by $B \rightarrow S + B$
(10)	0 2 7 7 8	aaaB	+ + \$	reduce by $S \rightarrow aB$
(11)	0 2 7 9	aaS	+ + \$	shift
(12)	0 2 7 9 10	aaS +	+ \$	reduce by $B \rightarrow \epsilon$
(13)	0 2 7 9 10 11	aaS + B	+ \$	reduce by $B \rightarrow S + B$

(14)	0 2 7 8	aaB	+\$	reduce by $S \rightarrow aB$
(15)	0 2 9	aS	+\$	shift
(16)	0 2 9 10	$aS +$	\$	reduce by $B \rightarrow \varepsilon$
(17)	0 2 9 10 11	$aS + B$	\$	reduce by $B \rightarrow S + B$
(18)	0 2 8	aB	\$	reduce by $S \rightarrow aB$
(19)	0 1	S	\$	shift

2 Optional Exercises

Exercise 3

1、There are 5 pairs of sets of items that can be merged.

$I_{2,7}$:

$S \rightarrow a \cdot B,$ +/\$
 $S \rightarrow \cdot S + B,$ +/\$
 $B \rightarrow \cdot,$ +/\$
 $S \rightarrow \cdot aB,$ +/\$

$I_{3,8}$:

$S \rightarrow aB \cdot,$ +/\$

$I_{3,8}$:

$S \rightarrow S \cdot + B,$ +/\$

$I_{5,10}$:

$B \rightarrow S + \cdot B,$ +/\$
 $B \rightarrow \cdot S + B,$ +/\$
 $B \rightarrow \cdot,$ +/\$
 $S \rightarrow \cdot aB,$ +/\$

$I_{6,11}$:

$B \rightarrow S + B \cdot,$ +/\$

The $LALR(1)$ parsing table for G' .

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2,7			1	
1			acc		

2,7	s2,7	r3	r3	4,9	3,8
3,8		r1	r1		
4,9		s5,10			
5,10	s2,7	r3	r3	4,9	6,11
6,11		r2	r2		

2、Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	<i>shift</i>
(2)	0 2,7	<i>a</i>	aaa + + + \$	<i>shift</i>
(3)	0 2,7 2,7	<i>aa</i>	aa + + + \$	<i>shift</i>
(4)	0 2,7 2,7 2,7	<i>aaa</i>	a + + + \$	<i>shift</i>
(5)	0 2,7 2,7 2,7 2,7	<i>aaaa</i>	+ + + \$	<i>reduce by $B \rightarrow \varepsilon$</i>
(6)	0 2,7 2,7 2,7 2,7 3,8	<i>aaaaB</i>	+ + + \$	<i>reduce by $S \rightarrow aB$</i>
(7)	0 2,7 2,7 2,7 4,9	<i>aaaS</i>	+ + + \$	<i>shift</i>
(8)	0 2,7 2,7 2,7 4,9 5,10	<i>aaaS +</i>	+ + \$	<i>reduce by $B \rightarrow \varepsilon$</i>
(9)	0 2,7 2,7 2,7 4,9 5,10 6,11	<i>aaaS + B</i>	+ + \$	<i>reduce by $B \rightarrow S + B$</i>
(10)	0 2,7 2,7 2,7 3,8	<i>aaaB</i>	+ + \$	<i>reduce by $S \rightarrow aB$</i>
(11)	0 2,7 2,7 4,9	<i>aaS</i>	+ + \$	<i>shift</i>
(12)	0 2,7 2,7 4,9 5,10	<i>aaS +</i>	+\$	<i>reduce by $B \rightarrow \varepsilon$</i>
(13)	0 2,7 2,7 4,9 5,10 6,11	<i>aaS + B</i>	+\$	<i>reduce by $B \rightarrow S + B$</i>
(14)	0 2,7 2,7 3,8	<i>aaB</i>	+\$	<i>reduce by $S \rightarrow aB$</i>
(15)	0 2,7 4,9	<i>aS</i>	+\$	<i>shift</i>
(16)	0 2,7 4,9 5,10	<i>aS +</i>	\$	<i>reduce by $B \rightarrow \varepsilon$</i>
(17)	0 2,7 4,9 5,10 6,11	<i>aS + B</i>	\$	<i>reduce by $B \rightarrow S + B$</i>
(18)	0 2,7 3,8	<i>aB</i>	\$	<i>reduce by $S \rightarrow aB$</i>
(19)	0 1	<i>S</i>	\$	<i>shift</i>